

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Area and volume

02

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

What is the length of one side of a square that has the same area as a circle with radius 2 ?

- A. 2
- B. $\sqrt{2\pi}$
- C. $2\sqrt{\pi}$
- D. 2π

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

A circle has a radius of 2.1 inches. The area of the circle is $b\pi$ square inches, where b is a constant. What is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Circle N has a radius of 7 millimeters mm. Circle M has an area of $64\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles N and M?

- A. 113π
- B. 92π
- C. 78π
- D. 15π

Circle K has a radius of 4 millimeters mm. Circle L has an area of 100π mm². What is the total area, in mm², of circles K and L ?

- A. 14π
- B. 28π
- C. 56π
- D. 116π

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

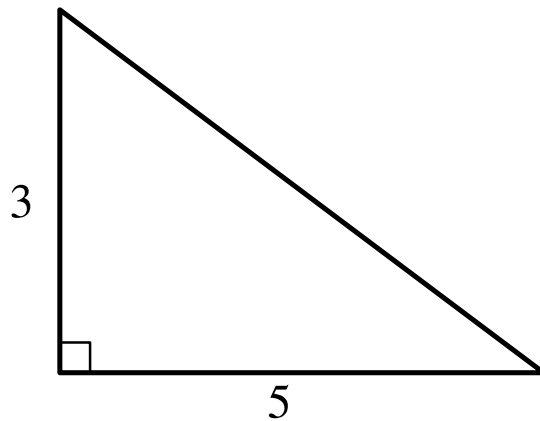
- A. 700
- B. 350
- C. 175
- D. 80

A right circular cylinder has a height of 8 meters m and a base with a radius of 12 m. What is the volume, in m^3 , of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

A cylindrical can containing pieces of fruit is filled to the top with syrup before being sealed. The base of the can has an area of 75 cm^2 , and the height of the can is 10 cm. If 110 cm^3 of syrup is needed to fill the can to the top, which of the following is closest to the total volume of the pieces of fruit in the can?

- A. 7.5 cm^3
- B. 185 cm^3
- C. 640 cm^3
- D. 750 cm^3



Note: Figure not drawn to scale.

- One angle is a right angle.
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 3
 - 5
- A note indicates the figure is not drawn to scale.

The figure shows the lengths, in inches, of two sides of a right triangle. What is the area of the triangle, in square inches?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

$\overline{9}$ $\overline{9}$ $\overline{9}$ $\overline{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A right circular cylinder has a base diameter of 22 centimeters and a height of 6 centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. 132π
- B. 264π
- C. 726π
- D. $2,904\pi$

The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2 , of the square?

- A. $2 \cdot 94$
- B. $2 \cdot 94 \cdot 94$
- C. $4 \cdot 94$
- D. $94 \cdot 94$